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Changes in seedlings' composition and abundance following soil scarification and amendments in a northern hardwood forest

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ABSTRACT

A combination of global changes (e.g., soil acidification, invasive species, forest management, climate) impacts the regeneration dynamics of sugar maple-dominated stands in northeastern North America. More specifically, there are concerns about the regeneration of sugar maple and other high-value species due to the expansion of American beech and its bark disease, killing mature trees, compromising fiber supply and other ecosystem services. Here, we set up a field trial to study the effect of soil scarification, amendments (nitrogen fertilization, liming, and wood ash), and their interactions, on tree species establishment following understory vegetation removal in a beech-invaded sugar maple stand to promote the establishment of high-value species (i.e. sugar maple and yellow birch), to control beech regeneration, and help reduce the impacts of beech bark disease on stand development. After four years of monitoring, we found that maple-beech stands with well-drained soils were promising sites for the success of sugar maple and yellow birch establishment. Soil scarification had a limited effect on sugar maple and beech establishment while it enhanced that of yellow birch, pin cherry and red maple. The combination of soil scarification and amendments was more effective in promoting the recovery of yellow birch and pin cherry and controlling the recovery of beech and red maple than their separated effects. Lime amendment promoted sugar maple establishment without favouring competitors such as yellow birch, pin cherry and red maple, and limited beech establishment. Overall, our results suggest that combining soil scarification and amendments on targeted site types could help fine-tune tree establishment conditions in northern hardwood forests.

1. Introduction

The northern hardwood forest is a major biome of southeastern and southcentral Canada and extends south into the United States. It is dominated by deciduous tree species, mainly sugar maple (*Acer saccharum* Marshall), American beech (*Fagus grandifolia* Ehrh), yellow birch (*Betula alleghaniensis* Britton), and companion tree species such as red maple (*Acer rubrum* Armstrong), and pin cherry (*Prunus pensylvanica* L. f.). These trees provide significant ecological, socio-economic, and cultural services. Sugar maple and yellow birch are considered the most valuable tree species for the forest industry with a wide variety of uses (Cale et al. 2017).

Although sugar maple has historically been a dominant species in northern hardwood forests, recent studies suggest that its competitive advantage may be declining in some areas (Battles et al. 2019, Oswald et al. 2018) due to various drivers such as extreme climatic/meteorological events including drought and freeze–thaw events (Gavin et al. 2008, Payette et al. 1996), insects (Hartmann and Messier 2008), subtle ecological effects induced by extreme events related to global change (Nolet and Kneeshaw 2018), competition with other species (Direction de la recherche forestière 2017), Ca^{2+} depletion with forest harvestings (Adams et al. 2000, Watmough et al. 2003) and soil acidification (Sullivan et al. 2013). Soil acidification consists of base cation loss (Ca^{2+}, Mg^{2+}, K^{+}, Na^{+}), and the most sensitive soils are those with low cation exchange capacity (CEC) and moderately high base saturation. Many factors cause soil acidification, including acidic precipitation (H^{+} in rainfall), atmospheric deposition of acidifying gases or particles such as sulphur dioxide (SO_2), ammonia (NH_3), and nitric and hydrochloric acids (HNO_3, HCL), the application of acidifying fertilizers such as elemental sulphur (S), urea or ammonium (NH^{4+}) salts, nutrient uptake by vegetation and root exudates, and the mineralization of organic matter (Goulding 2016). Soil pH is considered critical for seedling establishment (germination, survival, and growth), one of the most important and vulnerable processes in plant life and forest dynamics

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(Holste et al. 2011). For instance, soil acidity related to acid deposition, amplified by calcium depletion induced by forest harvesting, alters foliar Ca and Mg nutrition (Collin et al. 2016) and negatively affects the early stages of sugar maple seedling establishment (Juice et al. 2006, Kobe et al. 2002, Long et al. 2009).

American beech, called thereafter "beech", has been shown to have an increasing competitive advantage over sugar maple in some regions, particularly those with more acidic soils or those that have experienced disturbances such as logging (Cleavitt et al. 2021, Roy and Nolet 2018). Thus, shifts in competitive dynamics may be influenced by a variety of factors, including natural succession patterns and changes in soil conditions. The ability of beech to outcompete sugar maple is an increasing issue, and soil acidification processes may have shifted the competitive dynamic between beech and sugar maple (Cleavitt et al. 2021, Roy and Nolet 2018). In contrast to sugar maple, beech performs well on acidic and poor soils (Cleavitt et al. 2021, Duchesne and Ouimet 2008, 2009, Duchesne et al. 2002, Hane 2003), has a high shade tolerance (Beaudet et al. 1999), has a multi-faceted regeneration strategy that includes mast seeding and root suckering (Bannon et al. 2015, Cleavitt et al. 2008). The beech bark disease represents an additional issue. It greatly reduces the economic and ecological value of the stand once present, while it does not prevent beech from establishing (St-Jean et al. 2021).

Herbicide treatments have proven most efficient in controlling understory beech (Horsley 1994, Nyland et al. 2006); however, this approach is not allowed in Quebec. Among the potential solutions to reduce/control the dominance/competitiveness of beech over maple and other hardwood species in Quebec's hardwood forests, there are 1) the removal of beech trees/saplings/seedlings (Nyland et al. 2006); 2) the creation of large gaps with partial cuts that increase light availability (Dracup and MacLean 2018, Poulson and Platt 1996), 3) the correction of soil acidity and enhancement of soil fertility using amendments (Reid and Watmough 2014); and finally, 4) the exposure of the mineral soil by scarifying the soil to facilitate seed germination and seedling survival and growth of some species such as yellow birch (Gastaldello et al. 2007a, Natural Resources Canada 2019).

The mechanical removal of beech can limit understory beech invasion in the short-term (Horsley 1994). However, root injuries during this treatment also promote new suckers, along with some stump sprouts, which often maintain or increase the density of understory beech (Nyland et al. 2006). Beech removal is also inappropriate for treating large areas because it involves high cost and efforts (Bell et al. 1997a, 1997b). This approach is suggested for small acreages with a moderate density of small-diameter saplings (Nyland et al. 2006).

Several empirical studies suggested that significant gaps and partial cutting providing greater light availability in the understory could promote sugar maple regeneration over beech (Dracup and MacLean 2018, Poulson and Platt 1996, Canham 1988). Although sugar maple and beech have a similar ecological niche, under limited light availability (e.g., small gap, single-tree selection cut), beech performs better than sugar maple (Poulson and Platt 1996, Collin et al. 2017).

Lime and wood ash amendments have been used for forest soil amelioration to correct soil acidity and improve the soil's physical, biological, and chemical conditions (Lundström et al. 2003, Nolet et al. 2015). Lime amendment increases Ca^{2+} concentration and ionic strength in soil solution, causing flocculation and, thus, an improvement in soil structure and hydraulic conductivity (Haynes and Naidu 1998). In wood ash, Ca^{2+} is usually present as $CaCO_3$, which generates a liming effect on soil (Lundström et al. 2003) with a pH ranging from 8.9 to 13.5 (Demeyer et al. 2001). Earlier studies have shown that sugar maple growth and survival increased significantly in response to Ca addition, whereas beech had no growth response (Kobe et al. 2002, Long et al. 2011). However, wood ash has a lower neutralizing capacity in acidic soil than lime (Reid and Watmough 2014, Augusto et al. 2008).

Some forests such as northern hardwood forests are also limited in nitrogen (Vadeboncoeur 2010). Experiments with nitrogen addition have shown significant decreases in abundance and growth of sugar maple seedlings in response to increased N availability through a process known as nitrogen saturation causing changes in soil chemistry, microorganisms, and plant communities (Patterson et al. 2012, Talhelm et al. 2013, Zaccherio and Finzi 2007). The drawback of the nitrogen fertilizer effect is that ammonium salts strongly acidify the soil through nitrification (Marschner 2012). High level of N can cause an imbalance in the availability of the other essential nutrients, such as calcium and magnesium that are very important for sugar maple growth and survival. Excess N can also lead to an increase in the growth of competing vegetation which can outcompete sugar maple seedlings for resources such as light, water and nutrients.

Soil scarification has been suggested to improve seedlings' establishment in forests by controlling competing vegetation and creating microsites that promote seed masting, seedlings' growth, and survival (Natural Resources Canada 2019). It consists in removing the surface organic layers to expose and loosen the surface mineral soil and to create a suitable seedbed with higher mineral soil temperatures, more light getting to the seedling, and a more balanced supply of nutrients, oxygen, and water. However, Bédard et al. (2022) reported that soil scarification coupled with beech control and cutting treatments did not promote sugar maple at the seedling stage. In fact, soil scarification also has potential adverse effects, such as frost heaving (Chantal et al. 2003) or soil acidification through nutrient leaching due to increased soil erosion (Piirainen et al. 2007).

Clear understanding on how the interaction among these input materials and practices affects sugar maple establishment in maple-beech forests is still missing. Here we conducted a field experiment to evaluate the effects of different soil amendments and soil scarification, following selection cut and understory vegetation removal, on seedling establishment in beech-invaded a sugar maple-dominated stand at Valcartier in eastern Canada. Our objective was to find a potential silvicultural tool that simultaneously promotes seedlings of high-value species (sugar maple, yellow birch) and limits beech seedling establishment following selection cut. Our research questions were:

1. Which treatments among soil scarification and amendments and their interactions simultaneously limit beech and promote sugar maple, yellow birch, pin cherry, or red maple re-establishments?
2. How soil scarification and amendments affect soil nutrient availability?
3. Which microsite drivers have the most influential effects on species re-establishment?

2. Material and methods

2.1. Study location

The study was performed in the Boreal shield ecozone of Eastern Canada at the Canadian Forces Base Valcartier, near Quebec City (46.91° N, 71.58° W) (Fig. 1). The mean minimum temperature is −18.8 °C (January), and the mean maximum temperature is 23.8 °C (July), with an average precipitation of 1460 mm (Canadian Climate Normals 1981–2010 Station Data – Riviere Verte Ouest). We established the trial in a mature ∼ 70-year-old sugar maple-dominated stand with yellow birch, beech, pin cherry, and red maple as companion species (Tables 1 & S2). The stand (∼24 ha) was affected by the beech-bark disease and showed evident signs of maple regeneration failure: a significant portion of ground cover without regeneration (33%), few maple seedlings (<20%) and a high density of beech saplings (50%). The stand was subjected to a selection cut in 2014, which removed most of the beech trees and left 65% of the initial forest cover. The stand developed on sandy loam till material originating from granitic/gneiss formations of the Canadian Shield, and the soil is classified as Dystric Brunisols (orders Inceptisols in US soil taxonomy and Cambisols in the World Reference Base).

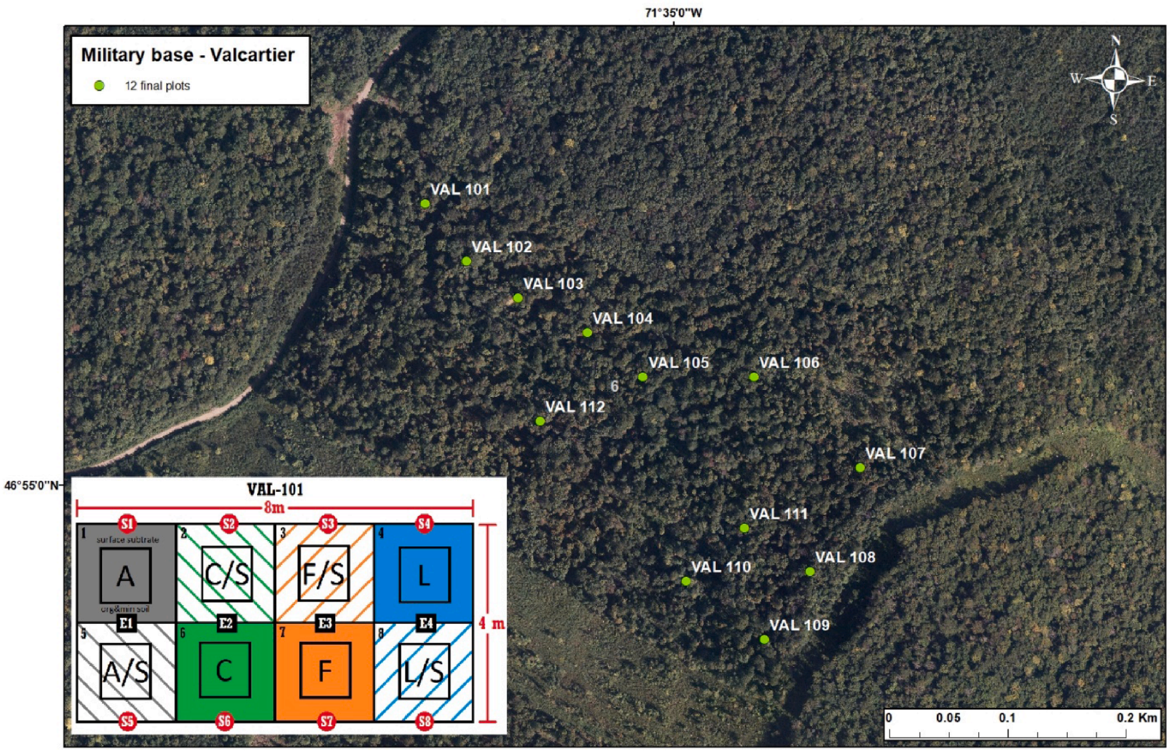

Fig. 1. Map of the study site showing the location of plots within the sugar maple-dominated stand in Valcartier. An example of a plot layout is represented in the lower-left corner (A = Ash; C = Control; F = Fertilizer; L = Lime; S = Scarification). A plot consisted in 8 subplots where the different treatments were applied. Each subplot included a buffer strip of 0.5 m around the measurement area.

Table 1
Stand characteristics before selection cutting and mineral soil characteristics at the beginning of the experiment (C = carbon; N = nitrogen; C.E.C. = cation exchange capacity).

Species	Density (stem/ha)	Volume (m^3/ha)	Basal area (m^2/ha)	Mean DBH (cm)	Seedling (% cover)	Mineral soil (mean ± sd)				
						C (%)	N (%)	C/N	pH H_2O	C.E.C. (cmol (+)/kg)
Gray birch (*Betula populifolia* Marshall)	8.8	1.7	0.3	22.0	–	6.34 ±2.40	0.34 ± 0.14	19.1 ± 2.05	4.49 ± 0.27	18.78 ± 6.24
Yellow birch (*Betula alleghaniensis* Britt)	108.8	36.6	5.0	24.3	3.3					
Red maple (*Acer rubrum* L.)	26.3	12.6	1.7	28.4	10					
Sugar maple (*Acer saccharum* Marsh.)	461.6	85.4	13.3	19.2	6.6					
Beech (*Fagus grandifolia* Ehrh)	53.6	25.7	3.0	26.7	50					
Trembling aspen (*Populus tremuloides* Michx.)	14.6	9.6	1.0	29.6	–					
Total	672.9	171.6	24.3							

3. Experimental design

In September/October 2017, before the establishment of a randomized blocks field trial using two levels of scarification (control and scarification) and four levels of amendments (control, lime, ash, and fertilizer), we cleared the understory cover (mainly beech saplings) and removed all woody debris (>1 cm diam). We laid out 12 plots (8 × 4 m each) containing four paired subplots (2 × 2 m each). We randomly assigned soil scarification to one subplot in each column, and the remaining subplot did not receive scarification (control). We assigned amendment treatments (lime, wood ash, fertilizer, and control) to each column. The control amendment was a subplot without any amendment (no lime, no wood ash, and no fertilizer). In total, one plot consisted in eight subplots with four scarified, four non-scarified, two amended with lime, two with ash, two with fertilizer and two with no amendment (Fig. 1).

To assess the effect of scarification on soil properties and species re-establishment, we lightly scarified four subplots in each plot in May 2018 with a particular toothed disk (UGS: 07 TS) mounted on a brush cutter (STIHL FS 361C-EM) to disturb the soil surface to a shallow depth

(1–2 cm), mix the organic layer with the mineral soil and partly to expose the mineral soil. We applied lime (0.5 kg/m^2 of $CaCO_3$), wood ash (0.885 kg/m^2), and fertilizer (0.5 kg/m^2 of NH_4NO_3) to one non-scarified subplot within each bock to evaluate the effect of these amendments alone on soil amelioration and species re-establishment. We applied lime, wood ash, and fertilizer to one scarified subplot within each plot to evaluate the effect of their interaction with soil scarification on soil amelioration and species re-establishment. We used one scarified and non-scarified subplot that did not receive any amendment within each plot to assess the effect of soil scarification alone.

The ash used in the trial was bottom ash collected from a nearby industrial biomass boiler (Resolute Forest Products, Château-Richer) that uses mainly conifer bark as feedstock (see Table S2 for ash chemical properties). The lime (powdered limestone, Mosher Limestone, Upper Musquodoboit, NS) and fertilizer (ammonium nitrate ≥ 95% ACS, Thermo Scientific Chemicals) were from certified providers of soil amendments.

4. Data collection

Data on biotic and abiotic variables were collected during the study period from September 2017 to October 2021. The biotic variables refer to woody plant's species abundance while the abiotic variables refer to soil properties and light availability.

Prior to site preparation, the number of woody species individuals was counted and classified by species and growth stage (seedling, sapling, and tree). These variables were coded as follow:

- Seedling: species name_Reg (e.g. *A.saccharum*_Reg);
- Sapling: Species name_shrub (e.g. *A.saccharum*_Shrub);
- Tree: Species name_Tree (e.g. *A.saccharum*_Tree).

We collected soil cores (0–20 cm) from the margin of each subplot to determine standard soil properties including total C and N, cation exchange capacity (CEC), pH and soil texture. Total C and N concentrations (%) were determined by dry combustion using a TruMac CNS analyzer (LECO Corporation, St. Joseph, MI, USA). The pH was determined in water using a 1:2 soil:water ratio. Exchangeable cations (as cmol+/kg) were determined by ICP-OES (Optima 7300 DV, Perkin Elmer, Waltham, MA, USA) following a Mehlich III extraction. The CEC was calculated as the sum of exchangeable cations in equivalent (Eq) unit (Carter and Gregorich 2007). Soil texture (sand, silt, and clay) was determined following the hydrometer method (Kroetsch and Wang 2007). Surface substrate type (organic, mineral, buried wood, stony) was recorded following soil core removal.

Monitoring of treatment effects on tree species re-establishment and soil chemistry started after treatment application in 2018. The term "seedling" refers to woody plantlet irrespective of the regeneration mechanism (i.e., seed, shoot dieback, sucker, or sprout origin). Seedling inventory was conducted twice during the growing season over four years (from 2018 to 2021), at the beginning (June) and end (October), except in 2020, where the inventory occurred only once and in August due to Covid-19 related restrictions. During each inventory period, the number of tree seedlings was counted and classified by species and height classes (0–15, 15–60, 61–130, and >130 cm) because we were interested in the process of regeneration from the beginning of the experiment to the end, and how the replacement is made between height classes and between species. Our intention was to assign different weights to a seedling of 10 cm and a seedling of 20 cm, as we did not want them to be considered equally important since competition for light resource is size-dependent (Bebre et al. 2021), a taller seedling has a competitive advantage over its smaller neighbors (Weiner 1990; Bauer et al. 2004).

Soil nutrient availability following treatments was evaluated in each subplot with ion-exchange resins (Ionac NM-60, Lenntech, Delft, The Netherlands; H+/OH − Form, Type I, Bead) during the growing seasons of 2018 and 2019 using the method outlined in De Grandpré et al. (2022). Briefly, the resins were placed in nylon bags, charged with 1 M HCl, and rinsed with deionised water. The bags were carefully inserted to a depth of 10 cm in the mineral soil in early June and recovered in mid-October. The resins were then rinsed with deionised water and separated into two samples. The first sample was extracted for 1 h in 2 M HCl, and the extract was filtered and analysed for P and exchangeable cations (summed) by ICP-OES (Optima 7300 DV, Perkin Elmer, Waltham, MA, USA). The cation exchange capacity (CEC) was calculated as the sum of exchangeable cations in equivalent (Eq) unit (Carter and Gregorich, 2007). The second sample was extracted for 30 min in 50 mL of 2 M KCl to determine NH4 + and NO3– by FIA (QuickChem 8500, Lachat Instruments, Loveland, CO, USA).

To evaluate the light availability reaching the understory strata, photosynthetic photon flux density (PPFD) measurements were conducted 4–5 times over the growing season using Li-200R pyranometers (Li-Cor, Nebraska, USA). Measurements were carried out at the center of each subplot while continuous measurements were performed at an adjacent opening to record the above canopy PPFD using a Li-1500 datalogger. The light availability for each subplot was then determined as %PPFD. Soil moisture measurements were conducted at the same time as these of light availability using a Fieldscout TDR 150 soil moisture meter.

4.1. Data analysis

We performed all analyses and graphs within the R statistical and programming environment version 4.2.0 (Team R Core 2021).

5. Data exploration

We conducted data exploration using the protocol described by Zuur et al. (2010). This protocol consists in checking outliers of dependent and explanatory variables, homogeneity of dependent variables, normality of dependent variables, zero trouble with dependent variables, collinearity of explanatory variables, relationships between dependent and explanatory variables, interaction, and independency of dependent variables). Exploring dependent variables (seedling abundance for each species and nutrient sorption by buried resin) using the "fitdistrplus" package (Delignette-Muller and Dutang 2015) revealed that none had a normal distribution. We investigated the presence of collinearity between explanatory variables using variance inflation (VIF). None of the variables had VIF higher than 2.5, meaning no collinearity among the explanatory variables (Zuur et al. 2007).

6. Effects of soil scarification and amendments on seedling establishment and soil nutrients

To assess the effect of soil scarification and amendment on species re-establishment and soil nutrient availability, we started by investigating what theoretical distribution came closest to the marginal distribution of the seedling abundance of each species and the abundance of each nutrient. We used the *fitDist ()* function of the "gamlss" package (Stasinopoulos et al. 2022) to find the best fit (based on the AIC of the different fitting attempts) among a list of available distributions adapted to count and positive real data in the "gamlss.dist" package (Stasinopoulos et al. 2018). Given the results (Table S3&4), we focused on distributions with the lowest AIC for each species. To test the effect of soil scarification and amendment on each species' re-establishment and nutrient contents, we used a Generalized Additive Model for Location, Scale, and Shape (GAMLSS) with the family distribution mentioned above for each response variable. GAMLSS family distributions have up to four parameters (μ, σ, ν, τ). The terms' location, scale, and shape refer to these parameters and are connected, but not necessarily equal, to the four moments of a distribution, namely mean (μ), variance (σ), skewness

(ν), and kurtosis (τ). GAMLSS is a modern distribution-based approach to regression analysis that expands the traditional approach to accommodate distribution parameters modelled as additive functions of predictor variables (Barber 2018). The fixed terms of our model were treatment (scarification and amendment), sampling period, seedling height class, and their two-and three-way interactions, whereas the random term was the plot. Model assumptions were verified by plotting residuals against fitted values to assess homoscedasticity, making QQ plots and computing filliben correlation coefficients to verify normality, and using worm plots to visualize model fit to the data (Stasinopoulos et al. 2017). A likelihood ratio test (LRT) was conducted to evaluate the influence of the explanatory variables in the models. When the p-value of a given term was < 0.05, the term significantly influenced species re-establishment.

7. The importance of microsite drivers in seedling establishment

We used a Random Forest analysis to find the most important drivers of species re-establishment from a set of microsites variables. This analysis models the averaging from sub-samples of observations and variables (Breiman 2003). The biotic predictor variables include the initial woody plant abundance (B.alleghaniensis_reg, B.papyrifera_reg, A.spicatum_reg, A.pensylvanicum_reg, A.rubrum_reg, A.saccharum_reg, F.grandifolia_reg, A.balsamea_reg, B.alleghaniensis_Shrub, A.pensylvanicum_Shrub, A.rubrum_Shrub, A.saccharum_Shrub, Deciduous. sp_Shrub, F.grandifolia_Shrub, A.pensylvanicum_Tree, A.balsamea_Tree), the covariate regeneration or specific seedling abundance during the experiment (B.alleghaniensis, B.papyrifera, A.spicatum, A.pensylvanicum, A.rubrum, A.saccharum, F.grandifolia, A.balsamea, V.alnifolium) and the species heigh class. The biotic variables can significantly influence the species re-establishment by inducing regeneration via sprout, sucker and seed rain (initial wood plant abundance) or by affecting resources availability (e.g. nutrients, light, water) and plant interactions (e.g. competition, facilitation, allelopathy, pathogens). The abiotic predictor variables include pre-treatment soil texture (sand, clay, silt), soil class (sandy loam, loam), soil strusbtrate (buried_wood, Organic, stony, mineral) and post-treatment soil properties (Soil_moisture_2018, Soil_moisture_2019, Soil_moisture_2020) and light availability (Light_2018, Light_2019, Light_2020). The time since treatment (period) was included also as predictor. The response variables were the seedling abundance of beech, sugar maple, red maple, yellow birch, and pin cherry. Numerous trees are built via a tree classification algorithm based on sampling the predictors and the observations. The prediction model is formed by the average prediction of all the regression trees forming the "forest". Each tree is grown by using the classification tree methodology without pruning. We used the "randomforest" package in R (Liaw and Wiener 2002) to quantify the importance of independent variables. The default value of "mtry" (the number of variables available for splitting at each tree node, here equal to the square root of the number of predictor variables) was used because of the conflicting evidence reported in the literature about the influence of this variable (Cutler et al. 2007, Strobl et al. 2008). However, due to the random way training data and variables are selected to determine the split at each node in Random Forest, importance ranking differs from one run to another (Millard and Richardson 2015). To deal with the variation of the importance value, we run a model with 1000 trees (Behnamian et al. 2017). We used the mean decrease in accuracy (%IncMSE) to measure predictor influence. %IncMSE is the average increase in squared residuals of the test set when the variable is pruned; the higher the value is, the higher the variable contribution to the overall variance of the predicted variable. This performance indicator is drawn separately for each tree of the forest and finally averaged over all trees of the "forest". In our analysis, we preferred %IncMSE over the indicator on the increase node purity (IncNodePurity) based on the Gini Index because it is the most robust and informative measure, while IncNodePurity is biased and should only be used if the extra computation time of calculating %IncMSE is unacceptable (Welling 2017). Variables with %IncMSE ≤ 0 were excluded in the graphs. We use the *ggpairs()* function from the "GGally" package in r (Schloerke et al. 2021) to find the relationship between the most influential variables and seedling abundance.

8. Results

8.1. Effect of soil scarification and amendment on species re-establishment

The seedling abundance of yellow birch was about 400 times higher than those of beech, pin cherry, sugar, and red maples. The models explaining species re-establishment as a function of soil treatments over time contain the main terms Treatment, time since treatment, seedlings height class, and 2- and 3- way interaction terms (Table 2). The models mean that the applied soil treatments had significant effects (positive or negative) on each species re-establishment, and the treatment effects varied (increased or decreased) significantly over time depending on the seedling height class. The models' validations indicated no violation of assumptions (i.e., independence and absence of residual patterns), and the worm plots show appropriate model fits (i.e., 95% of the worm plot lies within the dashed confidence bands; Figures S1-S5).

Sugar maple re-establishment significantly increased over time after

Table 2
Likelihood ratio tests (LRT) for the predictors of the GAMLSS model for μ term describing seedling abundance.

	df	LRT	Pr (Chi)
A. Sugar maple (BNB)			
Sampling period	5	151.69	<0.001
Height_class	3	513.03	<0.001
Treatment	7	62.40	<0.001
Random effect (plot)	11	2015.06	<0.001
Sampling period:Height_class	15	701.35	<0.001
Sampling period:Treatment	35	235.04	<0.001
Height_class:Treatment	21	590.86	<0.001
Sampling period:Height_class:Treatment	105	859.0	<0.001
B. Yellow birch (ZIBNB)			
Sampling period	5	38.03	<0.001
Height_class	3	974.60	<0.001
Treatment	7	197.30	<0.001
Random effect (plot)	11	147.842	<0.001
Sampling period:Height_class	15	1418.24	<0.001
Sampling period:Treatment	35	274.47	<0.001
Height_class:Treatment	21	1460.34	<0.001
Sampling period:Height_class:Treatment	105	2226.93	<0.001
C. Pin cherry (SI)			
Sampling period	5	106.40	<0.001
Height_class	3	769.52	<0.001
Treatment	7	160.31	<0.001
Random effect (plot)	11	66.290	<0.001
Sampling period:Height_class	15	1274.4	<0.001
Sampling period:Treatment	35	321.29	<0.001
Height_class:Treatment	21	1032.61	<0.001
Sampling period:Height_class:Treatment	105	1643.25	<0.001
D. Beech (ZIBNB)			
Sampling period	5	14.14	0.013
Height_class	3	352.46	<0.001
Treatment	7	17.39	0.015
Random effect (plot)	11	161.394	<0.001
Sampling period:Height_class	15	959.72	<0.001
Sampling period:Treatment	35	44.56	0.567
Height_class:Treatment	21	379.56	<0.001
Sampling period:Height_class:Treatment	105	1042.35	<0.001
E. Red maple (ZIBNB)			
Sampling period	5	63.16	<0.001
Height_class	3	454.41	<0.001
Treatment	7	92.85	<0.001
Random effect (plot)	11	473.09	<0.001
Sampling period:Height_class	15	697.30	<0.001
Sampling period:Treatment	35	102.13	<0.001
Height_class:Treatment	21	501.99	<0.001
Sampling period:Height_class:Treatment	105	795.76	<0.001

treatment. This re-establishment was less successful on scarified soil in the first two years (2018 and 2019), but this trend shifted later (Table 2A; Fig. 2). Lime and wood ash amendment significantly increased sugar maple re-establishment under scarified and non-scarified soils. The recruitment in higher height classes increased over time with more successful re-establishment on non-scarified soils under the wood ash amendment, followed by non-scarified soils under the lime amendment. Sugar maple had no dominant individuals (no seedling higher than 130 cm).

Yellow birch and pin cherry re-establishment (height $\geq$ 15 cm) decreased significantly over time with higher success (three to fourteen times in the first year) on scarified soil (Table 2B & C; Figs. 3 & 4). On scarified soil, the re-establishment was sometimes significantly more successful on unamended soil than on amended soil. The recruitment in higher height classes increased over time with more successful re-establishment on scarified soil. Pin cherry had no dominant individuals (no seedling higher than 130 cm).

Beech seedling establishment significantly decreased over time (Table 2D; Fig. 5). Sometimes this establishment was more successful on scarified soil under fertilizer and lime amendments. Most of the time, the re-establishment was less successful on scarified soil under the wood ash amendment. On non-scarified soils, the re-establishment was less successful under fertilizer and lime amendments while more successful under wood ash amendment. The recruitment in higher height classes increased over time with a clear effect of wood ash amendment in controlling re-establishment on scarified soil.

Red maple seedling establishment varied over time without a clear trend across soil treatments. Most of the time, seedling establishment was less successful on scarified soil and under lime amendment (Table 2E; Fig. 6). Ash amendment and fertilizer did not significantly improve seedling establishment. Most of the time, the recruitment in higher height classes was substantially higher on non-scarified soils.

9. Effect of soil scarification and amendment on soil nutrient

The first year following treatments (2018), wood ash addition resulted in a substantially higher increase in most soil nutrients relative to the control, more than lime and fertilizer, whereas the fertilizer increased mainly the nitrogen forms (Table 3; Fig. 7, Figures S6-S16). For some nutrients, scarification amplified this increase from two to five times. The second year after treatments, most soil nutrients under all amendments dropped drastically, almost at the same level. The sorption of Ca and Mg increased significantly under the lime amendment in the second year, while the sorption of Mn and Fe remained almost the same in the two years across the different amendments.

10. Drivers of species re-establishment

The random forest (RF) analysis revealed that the importance of biotic, abiotic and soil treatment predictors of the seedling abundance was species-dependent (Fig. 8). The microsite drivers explained 29.04% of sugar maple re-establishment. The most important driver of sugar maple seedling abundance was seedling height class followed by soil moisture in two years after treatments and the time since treatment (Fig. 8A), and seedling abundance increased significantly as soil moisture increased. The other microsite drivers had lower importance with

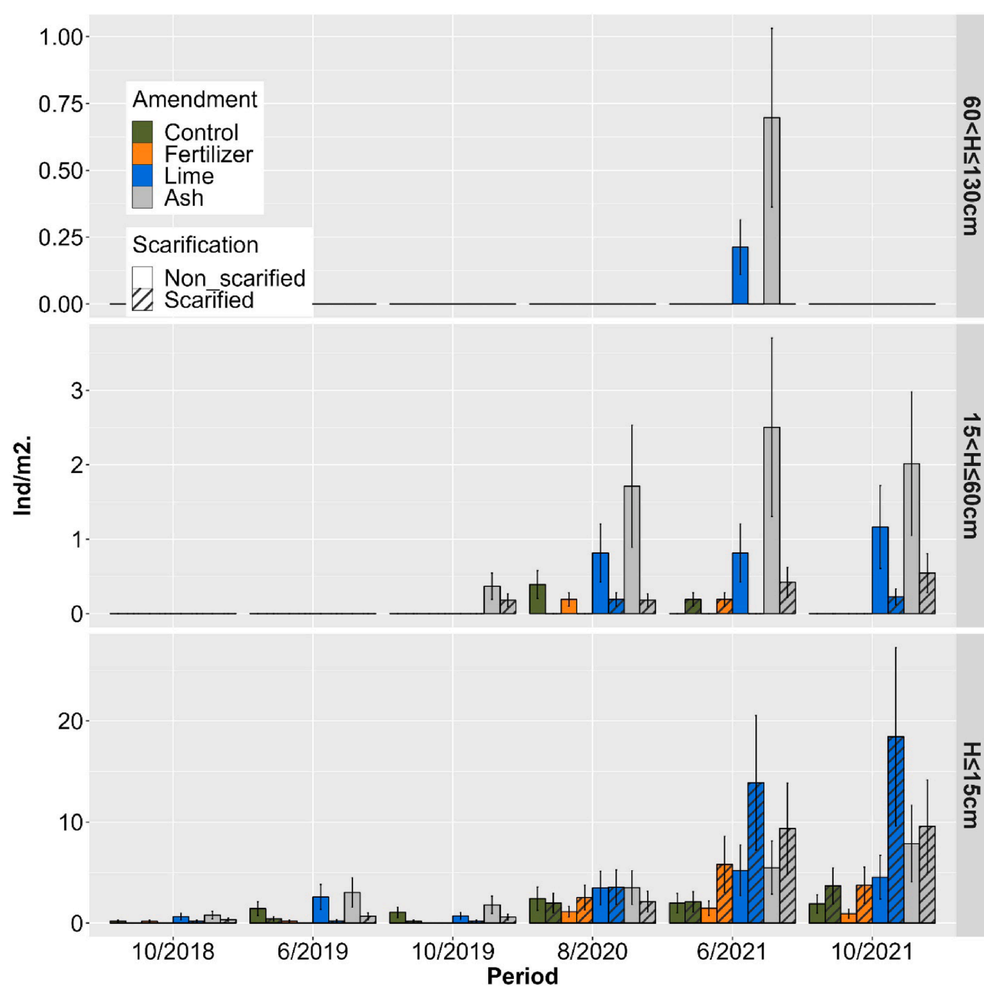

Fig. 2. Sugar maple re-establishment response (in individual per square meter) to soil treatments over four years for different height classes (H = Height).

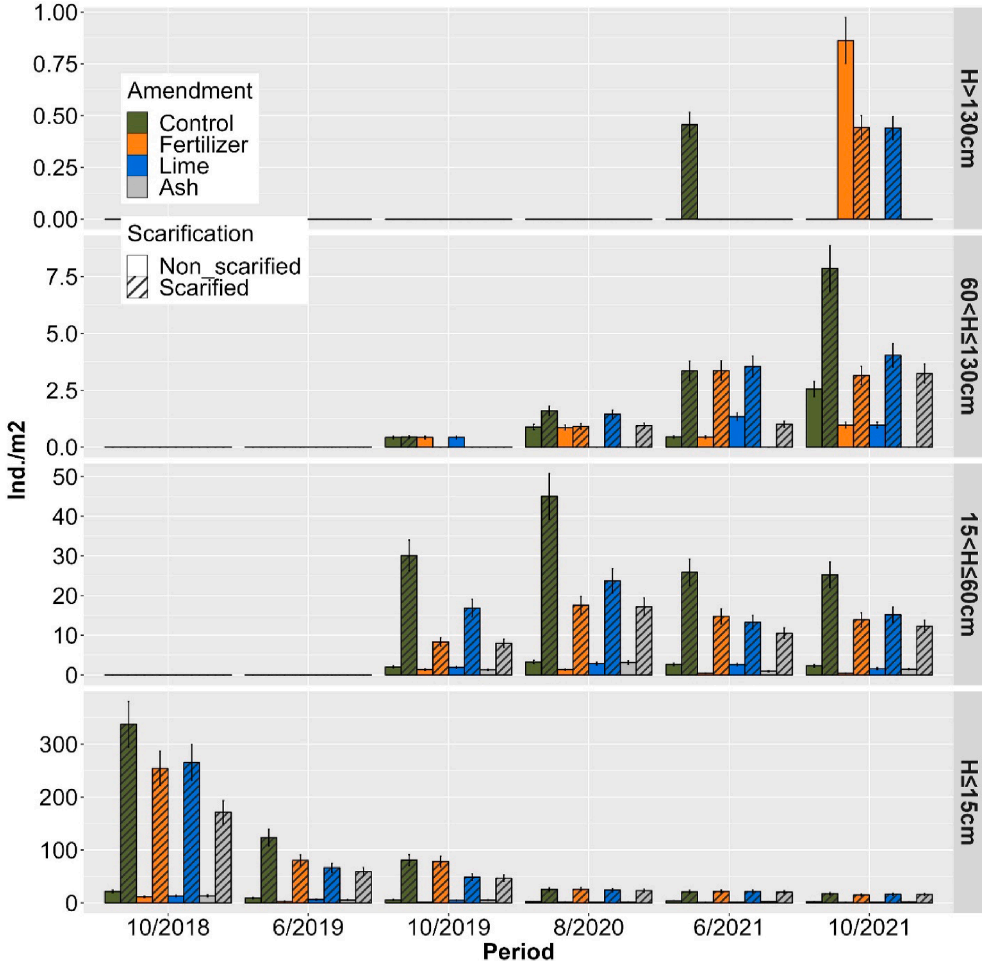

Fig. 3. Yellow birch re-establishment response (in individual per square meter) to soil treatments over four years for different height classes (H = Height). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

beech seedling and shrub abundance coming after red maple, yellow birch, and pin cherry seedlings abundance; light availability after soil texture; soil scarification after soil amendment and the interaction between scarification and amendment.

The microsite drivers explained 53.22% of yellow birch re-establishment. The most important driver was seedling height class followed by the time since treatment, the abundance of pin cherry seedlings, soil scarification and the interaction between scarification and amendment (Fig. 8B), and yellow birch seedling establishment increased significantly as the abundance of pin cherry seedlings increased. The other microsite drivers had lower importance with light availability coming before soil properties. Soil amendment alone was a poor driver of yellow birch re-establishment.

The microsites drivers explained 44.54% of beech re-establishment. The most important driver was yellow birch seedling abundance followed by light availability of the first year after treatment, pin cherry seedling abundance, seedling height class, and beech shrub abundance before treatment (Fig. 8C), and beech re-establishment increased significantly as both yellow birch and pin cherry seedling abundance and light availability increased. The other microsite drivers had lower importance with light availability of the second and third years coming after soil moisture, the combined effect of soil scarification and amendment before amendment and scarification.

The microsites explained 31.62% of red maple re-establishment. The most important driver was yellow birch seedling abundance followed by the seedling abundance of red maple before treatment, seedling height class, red maple shrub abundance before treatment, and light availability of the first year (Fig. 8D), and red maple re-establishment increased significantly as yellow birch seedling abundance, red maple seedling and shrub abundance before soil treatments increased. The other microsite drivers had lower importance with light availability coming before soil moisture and texture; soil scarification alone before soil amendment alone, and the interaction between amendment and scarification.

The microsite driver explained 55.48% of pin cherry re-establishment. The most important driver was the time since treatment followed by the abundance of yellow birch seedlings, seedling height class, light availability of the second year, the combined effect of soil scarification and amendment, and soil scarification (Fig. 8E), and pin cherry re-establishment increased significantly as the abundance of yellow birch seedlings increased. The other microsite drivers had lower importance with light availability coming before soil properties.

11. Discussion

11.1. Effect of soil scarification and amendment on soil nutrient and species re-establishment

Wood ash and lime amendments have been used as acidic-neutralizing materials to promote sugar maple re-establishment and control beech invasion and beech bark disease (Sullivan et al. 2013, Lawrence et al. 2018). In this study, we have shown how wood ash and lime amendments promote sugar maple re-establishment in northern hardwood forest, while fertilizer had no detectable effect. The large

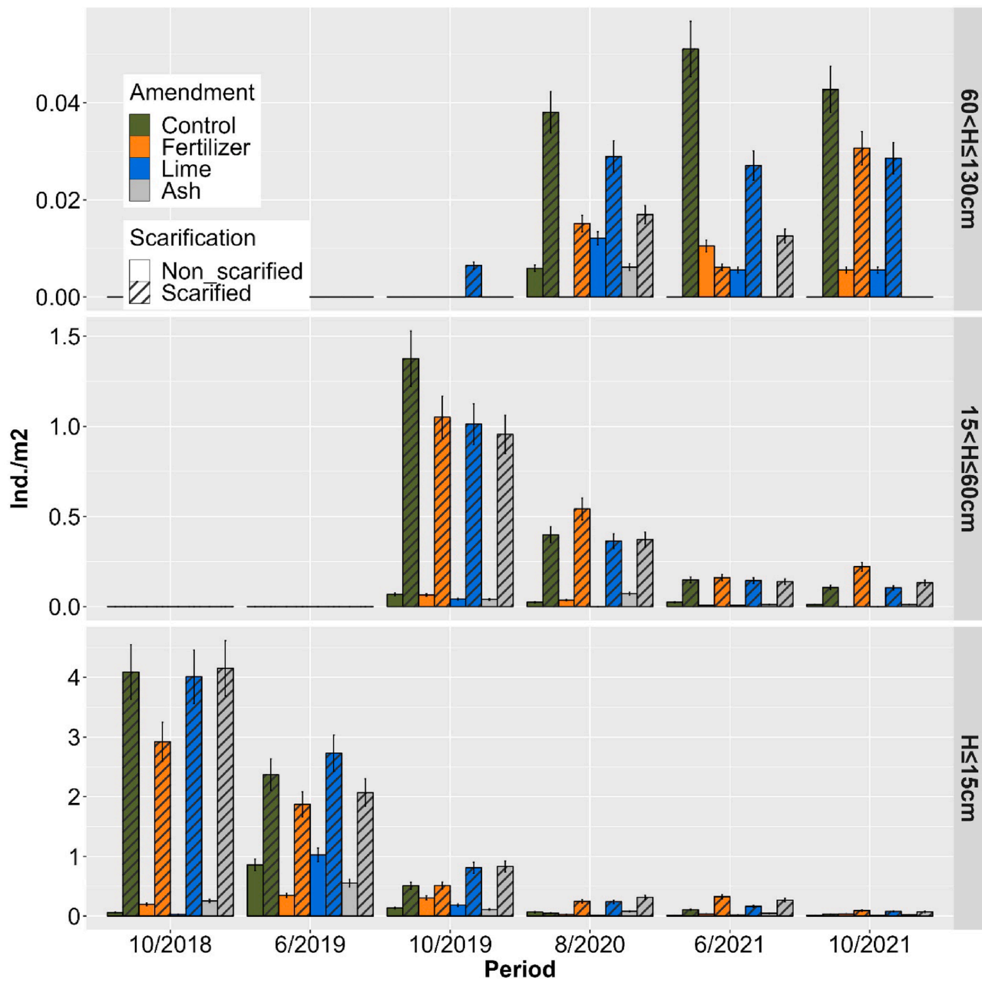

Fig. 4. Pin cherry re-establishment response (in individual per square meter) to soil treatments over four years for different height classes (H = Height).

increase in soil nutrients (particularly base cations) with ash and lime amendments may explain the response of sugar maple since the species is known to be sensitive to calcium depletion (Sullivan et al. 2013). In contrast, the substantial increase in nitrogen forms provided by the fertilizer may have caused soil acidification due to nitrification and nitrate leaching (Goulding 2016). This phenomenon may also explain the lower success of sugar maple re-establishment following wood ash amendment on scarified soils versus non-scarified soils: the soil scarification being associated with an increased availability of NH_4^+ and NO_3^- and the promotion of competitors such as yellow birch and pin cherry (Figs. 5 & 6). In addition, nitrogen fertilizer stimulates the germination of dormant pin cherry seed (Auchmoody 1979). Therefore, ammonium-based fertilization and soil scarification may not be appropriate to promote sugar maple re-establishment on acidified soils. Although the combination of soil scarification and ash amendment promoted sugar maple re-establishment and controlled beech invasion, soil scarification substantially enabled yellow birch and pin cherry re-establishment. Lime addition allowed to limit beech invasion and promote sugar maple re-establishment without promoting yellow birch, pin cherry and red maple re-establishments. This amendment is known to have a longer-term effect and a higher soil acidic neutralization power than wood ash (Pitman 2006).

The study results also reveal that sugar maple and pin cherry did not have dominant individuals compared to the other species. This finding suggests that the studied stands will be dominated by beech, yellow birch and red maple. However, since the results are from a short term and small-scale experiment, and focus on seedling establishment, they can't be extrapolated to sapling or adult tree stage. Because only one or two individuals of the seedlings should reach these stages within subplots after self-thinning and eventual browsing (see Picture S1). The promotion of a given species seedling by a treatment does not necessarily mean that these seedlings will be recruited at sapling stage, and in the opposite the limitation of seedling establishment does not also mean that these seedlings will not be recruited at sapling stage. For these reasons, we will pursue the survey until the original seedling reach sapling stage in order to get the full effect of the treatments on sapling and tree establishment.

12. Microsite drivers of species re-establishment

Microsite drivers, both biotic and abiotic, play important roles in the establishment of sugar maple, yellow birch, pin cherry and beech seedling re-establishment. We found that the most important microsite drivers were biotic variables such as seedling height class, yellow birch and pin cherry seedling abundance. This suggest that biotic factors such as competition or facilitation are among the most influential microsite drivers of seedling establishment. Sugar maple, yellow birch, pin cherry and beech seedlings compete with other individuals (both intra- and interspecific competition) for resources such as light, water and nutrients. However, sugar maple seedlings are more sensitive to competition than beech seedling. Studies have shown that sugar maple seedling are more likely to die in the presence of other vegetation, while beech seedlings can better tolerate competition and can even grow taller in the presence of other vegetation (Cleavitt et al. 2021, Roy and Nolet 2018). There may also be facilitation interactions between beech and both yellow birch and pin cherry, that may increase the competition pressure

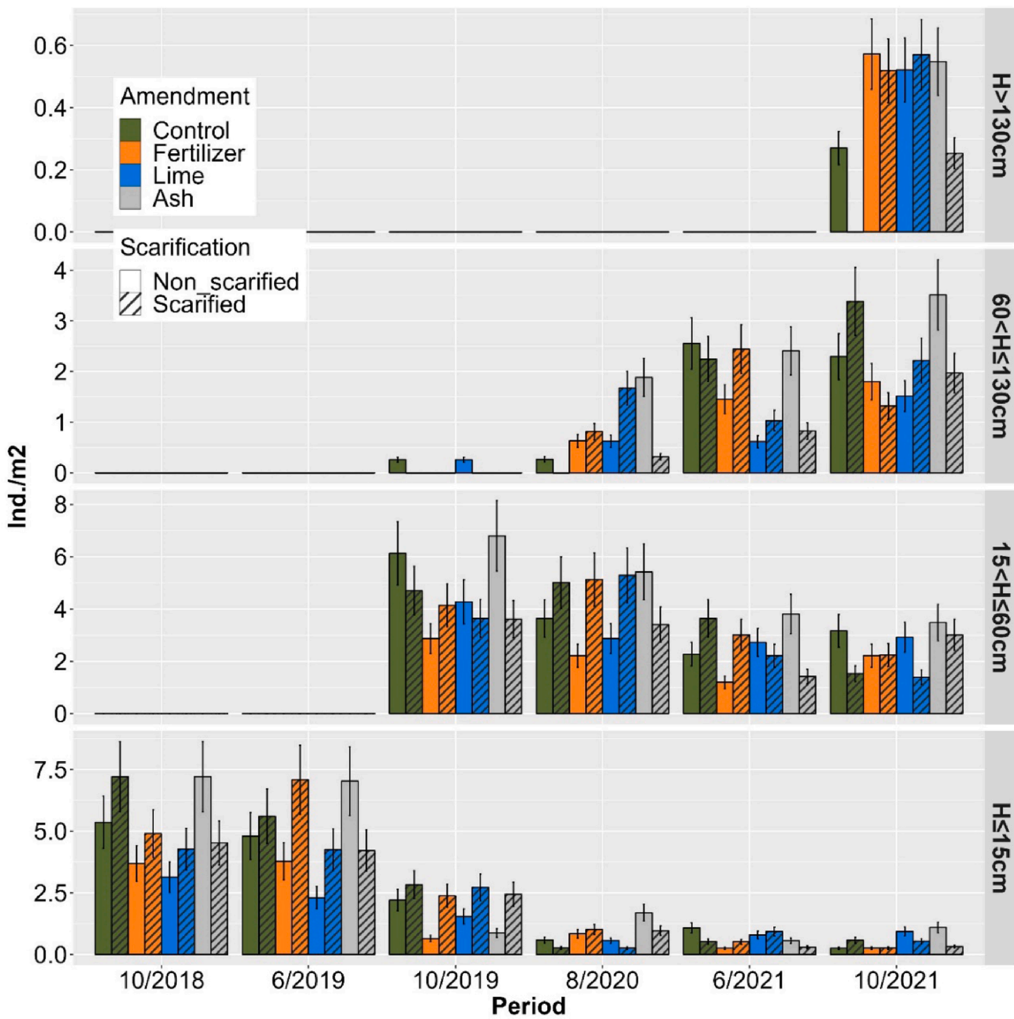

Fig. 5. Beech re-establishment response (in individual per square meter) to soil treatments over four years for different height classes (H = Height).

on sugar maple seedling. This suggests that competition with other species is a key factor influencing sugar maple seedling establishment.

Abiotic drivers such as soil moisture and light availability were among the most important microsite drivers of seedling re-establishment. Soil moisture represents an essential factor limiting seedling establishment because water uptake by seed is a prerequisite for proper germination and seedling establishment (Hadas A. 2005). Furthermore, the successful establishment of seedlings depends on the availability of nutrients (Leck et al. 2008). Both sugar maple and beech seedling require moist soils to establish and grow, but sugar maple seedlings are more sensitive to drought than beech seedlings. This is consistent with previous studies reporting that sugar maple prefer well-drained soils with ample moisture (Godman 1965), while beech does well on dry-mesic soils (Tubbs 1978). These findings suggest that northeastern hardwood forest stands with well-drained soils are promising sites for sugar maple re-establishment, growth, and dominance after selecting cut and soil preparation.

Light availability is a critical drivers of seedling establishment and growth. Although sugar maple and beech are shade-tolerant, they have different light requirements and tolerances. After selecting cut and site preparation, beech seedlings were more sensitive to light availability than sugar maple seedlings. This is consistent with previous studies reporting that small gaps in the overstory did not control beech establishment over sugar maple (St-Jean et al. 2021, Poulson and Platt 1996, Collin et al. 2017, Nolet et al. 2008). Besides the promotion of the seedling establishment of species (beech, red maple, pin cherry, yellow birch) that may over compete sugar maple, removing the overstory canopy may also reduce soil moisture, potentially impacting negatively sugar maple seedling survival. This finding suggests that increasing light availability through selection cutting and soil preparation, is not an efficient approach to simultaneously promote sugar maple and limit beech re-establishments.

Soil scarification was among the most influential drivers of yellow birch and pin cherry re-establishments. In contrast, sugar maple and beech re-establishments were less influenced by soil scarification, consistent with Thurston & Krasny (1992). Gastaldello et al. (2007b) reported that while yellow birch needed exposed mineral soil to establish, better growth and higher nutrient availability was found on the edge of exposed mineral soil patches or in mounds created by the removal of the forest floor. These findings suggest that soil scarification has limited effect on sugar maple and beech re-establishment compared to yellow birch, pin cherry and red maple in this northern hardwood forest.

Soil amendment was more influential than soil scarification for sugar maple and beech re-establishments, while the opposite was found for red maple, pin cherry and yellow birch. The interaction between soil scarification and soil amendment was more influential for sugar maple, beech and red maple than a single treatment, but its effect on yellow birch and pin cherry was similar to that of soil scarification. These findings are partly consistent with our expectation that soil scarification might amplify the lime, wood ash and fertilizer effects by increasing the receptivity of the germination beds, soil temperature and availability of

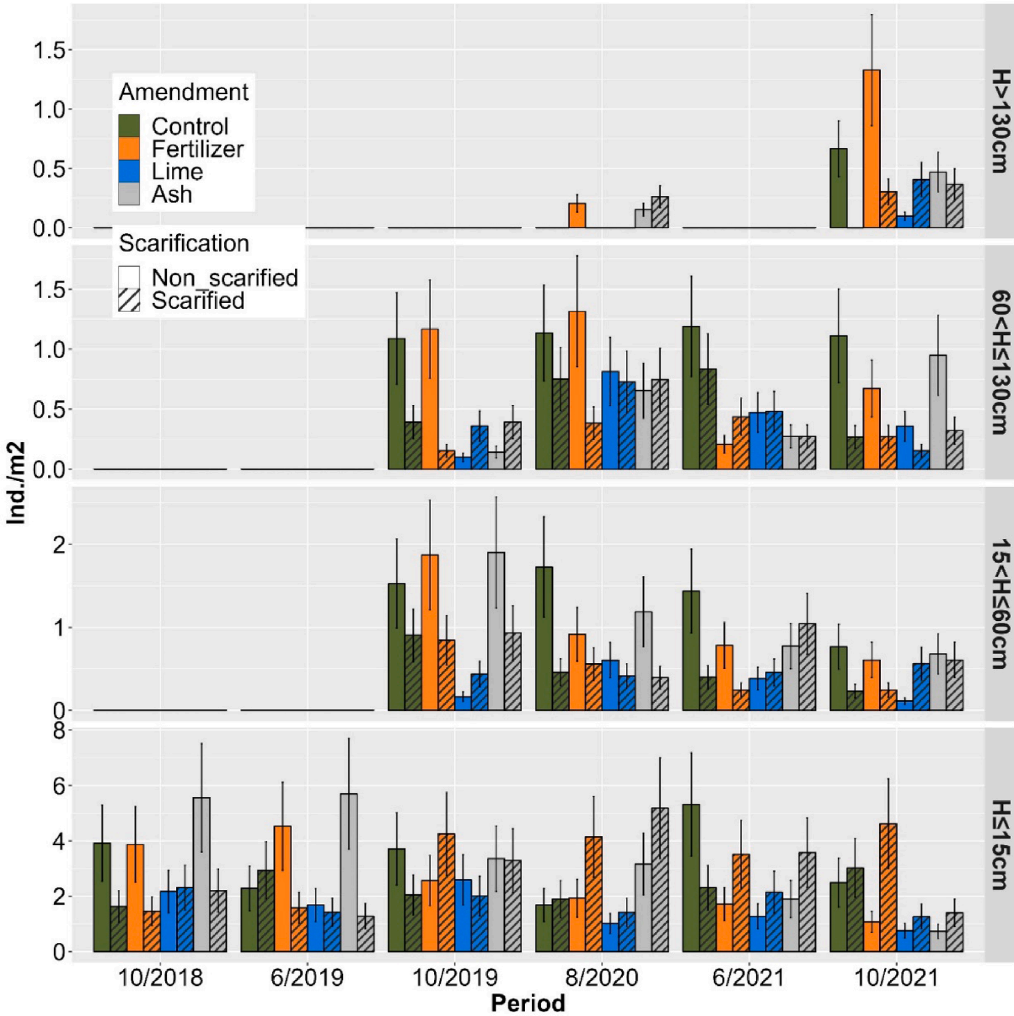

Fig. 6. Red maple seedling establishment response (in individual per square meter) to soil treatments over three years for different height classes (H = Height). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

resources (e.g., light, water, nutrients), setting initial control of competing vegetation and improving the establishment, survival and growth of seedlings. These findings suggest that soil amendment and soil scarification have different effects on the re-establishment of sugar maple, beech, yellow beech, red maple and pin cherry. Additionally, the interaction between treatments is more influential for some species than others, highlighting the importance of considering the specific needs of different when implementing site preparation to fine-tune forest composition.

It has to be noted that the success of seedling establishment during early successional stages results first from seed supply and dispersal, secondly from each species' specific germination, seedling (originate from both sexual and asexual reproduction, including resprouting from roots or stumps) survival, and growth requirement in resources such as water, nutrient and light. Therefore, the success of re-establishments of the studied species varies with their tolerance to resource availability.

13. Implications

The major drivers of the studied species re-establishment success or failure should dictate the alternatives for managing species promotion or control. This study exhibits 4 categories of drivers listed below. Amongst those, site factors play an important role and should be first considered before devoting too much efforts into managing for valued tree species. Second, specific actions including amendment and site preparation need to be evaluated carefully and monitored throughout stand development.

1. Abiotic factors that are difficult to manipulate; soil moisture is the most influential driver of sugar maple re-establishment. Sugar maple stands with well-drained soils are promising sites to consider in promoting the success of its re-establishment in northeastern hardwood forests.
2. Abiotic factors that can be modified by using common silvicultural practices; light availability is the most influential driver of beech re-establishment that increases as light availability increases in small gaps. Selecting cut and beech shrub removal are not efficient to limit the success of beech re-establishment in the studied stand.
3. Abiotic and biotic factors that can be modified by using soil preparation such as soil scarification, wood ash, and lime amendments. Lime amendment is an expensive practice (extraction, transportation, and field application). Its extraction is also associated with substantial atmospheric carbon dioxide emissions (European Commission 2001). For these reasons, the combination of soil scarification and ash amendment is an interesting alternative to limit the success of beech and promote sugar maple and yellow birch re-establishment. However, this combination of soil treatment substantially enables pin cherry re-establishment, a species of low economic value. Ash amendment is less expensive, and the forest industry generates large amounts of wood ash as a waste product.

Table 3
Likelihood ratio tests (LRT) for the predictors of the GAMLSS model for μ term describing soil properties.

	df	LRT	Pr (Chi)
A. N.NH4 (GG)			
Year	1	35.11	<0.001
Scarification	1	0.84	0.394
Amendment	3	7.43	0.049
Random effect	11	32.00	<0.001
Year: Scarification	1	37.09	<0.001
Year: Amendment	3	55.10	<0.001
Scarification: Amendment	3	11.86	0.108
Year: Scarification: Amendment	3	66.04	<0.001
B. N.NO3 (PARETO2)			
Year	1	37.01	<0.001
Scarification	1	0.69	0.411
Amendment	3	9.31	0.029
Random effect	11	52.55	<0.001
Year: Scarification	1	43.61	<0.001
Year: Amendment	3	62.73	<0.001
Scarification: Amendment	3	13.96	0.059
Year: Scarification: Amendment	3	77.32	<0.001
C. P (IG)			
Year	1	11.41	<0.001
Scarification	1	0.72	0.396
Amendment	3	70.13	<0.001
Random effect		35.87	<0.001
Year: Scarification	1	12.78	0.005
Year: Amendment	3	75.92	<0.001
Scarification: Amendment	3	78.87	<0.001
Year: Scarification: Amendment	3	88.57	<0.001
D. K (GG)			
Year	1	26.03	<0.001
Scarification	1	5.22	0.056
Amendment	3	41.71	<0.001
Random effect	11	13.67	<0.001
Year: Scarification	1	28.83	<0.001
Year: Amendment	3	146.15	<0.001
Scarification: Amendment	3	48.72	<0.001
Year: Scarification: Amendment	3	161.88	<0.001
E. Ca (LOGNO2)			
Year	1	0.07	0.786
Scarification	1	0.88	0.375
Amendment	3	109.02	<0.001
Random effect	11	24.77	<0.001
Year: Scarification	1	3.87	0.305
Year: Amendment	3	114.49	<0.001
Scarification: Amendment	3	113.86	<0.001
Year: Scarification: Amendment	3	127.38	<0.001
F. Mg (IG)			
Year	1	0.98	0.323
Scarification	1	0.38	0.534
Amendment	3	134.35	<0.001
Random effect	11	23.63	0.001
Year: Scarification	1	1.40	0.706
Year: Amendment	3	149.22	<0.001
Scarification: Amendment	3	136.72	<0.001
Year: Scarification: Amendment	3	160.50	<0.001
G. Mn (BEINF)			
Year	1	0.71	0.394
Scarification	1	2.95	0.093
Amendment	3	38.86	<0.001
Random effect	11	63.34	<0.001
Year: Scarification	1	5.97	0.116
Year: Amendment	3	50.56	<0.001
Scarification: Amendment	3	44.66	<0.001
Year: Scarification: Amendment	3	60.58	<0.001
H. Al (LOGNO2)			
Year	1	17.07	<0.001
Scarification	1	3.58	0.061
Amendment	3	2.57	0.471
Random effect	11	50.89	<0.001
Year: Scarification	1	22.04	<0.001
Year: Amendment	3	25.44	<0.001
Scarification: Amendment	3	12.33	0.093
Year: Scarification: Amendment	3	39.52	<0.001
I. Fe (GG)			
Year	1	2.89	0.103
Scarification	1	1.61	0.220
Amendment	3	8.30	0.042
Random effect	11	51.55	<0.001
Year: Scarification	1	4.66	0.222
Year: Amendment	3	14.67	0.046
Scarification: Amendment	3	18.79	0.009
Year: Scarification: Amendment	3	32.15	0.008
J. Na (GG)			
Year	1	22.22	<0.001
Scarification	1	<0.001	1
Amendment	3	35.97	<0.001
Random effect	11	77.67	<0.001
Year: Scarification	1	27.50	<0.001
Year: Amendment	3	203.53	<0.001
Scarification: Amendment	3	43.76	<0.001
Year: Scarification: Amendment	3	224.83	<0.001
K. C.E.C (LOGNO2)			
Year	1	7.757	0.004
Scarification	1	0.06	0.789
Amendment	3	89.47	<0.001
Random effect	11	24.94	<0.001
Year: Scarification	1	10.22	0.015
Year: Amendment	3	124.20	<0.001
Scarification: Amendment	3	90.16	<0.001
Year: Scarification: Amendment	3	131.97	<0.001

Also, there is a growing use of harvest residues as an energy source, which generates ashes that are often landfilled. The removal of beech can also provide an alternative feedstock for bioenergy production. This practice could control the species' abundance and provides wood ash that may be used as amendment to improve acidic forest soils.

4. Environmental drivers that are not considered in this experiment (i.e., unexplained variance in the random forest analysis), but are influential drivers of seedling establishment. The control of heavy herbivory should be considered in areas with extensive browsing pressure because herbivores such as deer primary avoid beech in favor of more palatable species (Nyland et al. 2016). The forest stand composition before abiotic and biotic factors manipulation is important in species re-establishment. The elimination of beech seed trees through cutting all the large trees or poisoning or girdling them to reduce the seed supply and limit subsequent root suckering (Jones et al. 1989; Houston 1997).

14. Conclusion

To fine-tune tree establishment following selecting cut and soil preparation in northern hardwood forests, moist and well-drained soils are promising sites for the success of re-establishing sugar maple and yellow birch. Soil scarification had limited effect on sugar maple and beech re-establishments compared to yellow birch, pin cherry and red maple. Creating small gaps in forest overstory by applying selection cutting did not appear as an efficient approach for promoting sugar maple and limiting beech re-establishments simultaneously. Combining soil scarification and amendment was more effective in promoting sugar maple and controlling beech, and red maple re-establishments. Lime amendment was an interesting alternative for limiting beech re-establishment and promoting sugar maple without promoting the establishment of competitors such as yellow birch, pin cherry and red maple. Overall, our results suggest that a combination of soil scarification and amendments could help fine-tune forest composition and dominance in high-value northern hardwood forests. Since these results apply to early stages of species re-establishment, longer-term monitoring is needed to demonstrate the applicability of these results to later stages of stand development.

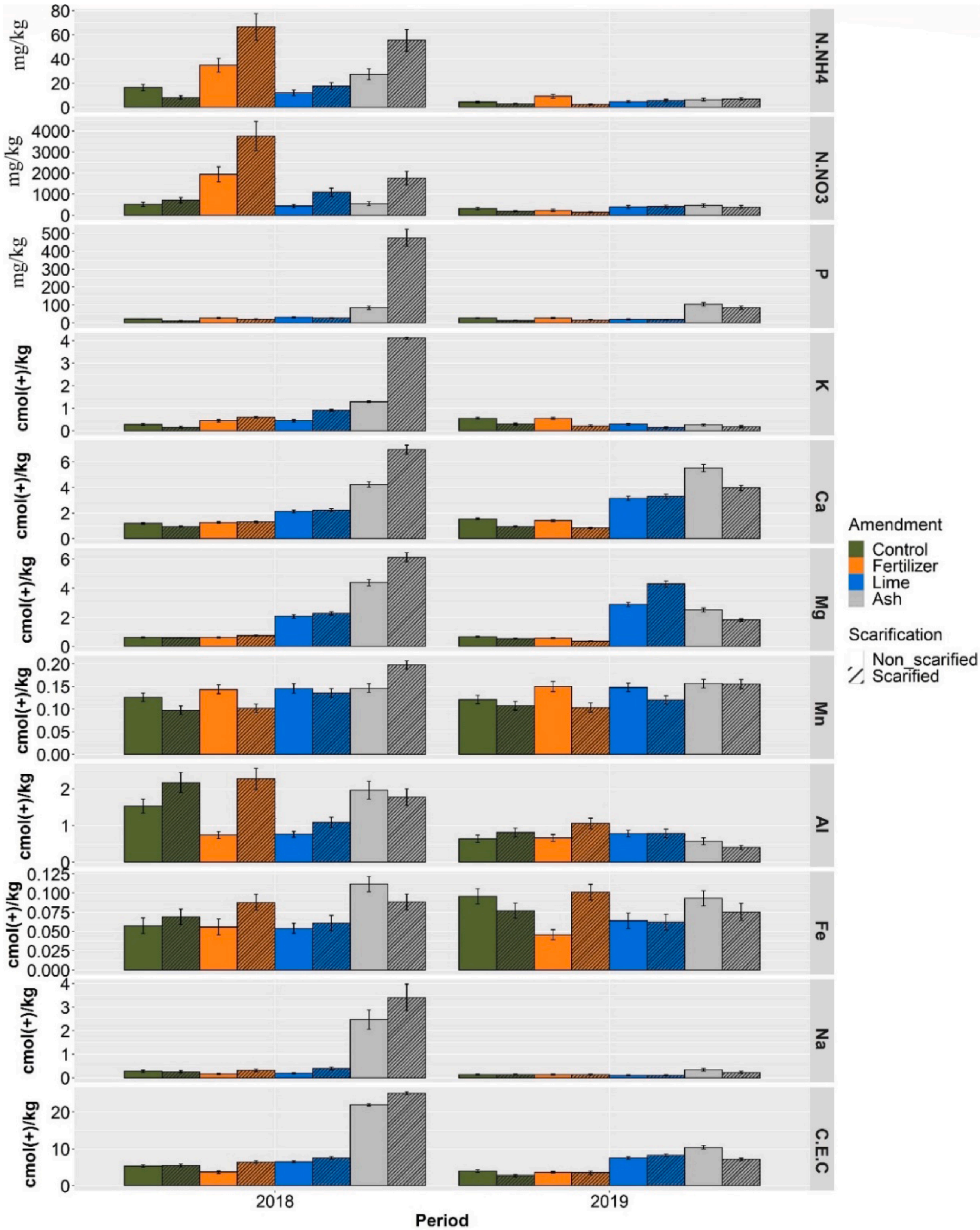

Fig. 7. The effect of soil treatments on soil nutrient properties response (N.NH4=; N.NO3; P = Phosphorus; K = Potassium; Ca = Calcium; Mg = Magnesium; Mn = Manganese; Al = Aluminum; Fe = Iron; Na = Sodium; C.E.C = Cation exchange capacity).

CRediT authorship contribution statement

Fidèle Bognounou: Visualization, Writing – original draft, Writing – review & editing. **David Paré:** Conceptualization, Validation, Writing – original draft, Writing – review & editing. **Jérôme Laganière:** Conceptualization, Funding acquisition, Project administration, Supervision, Methodology, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

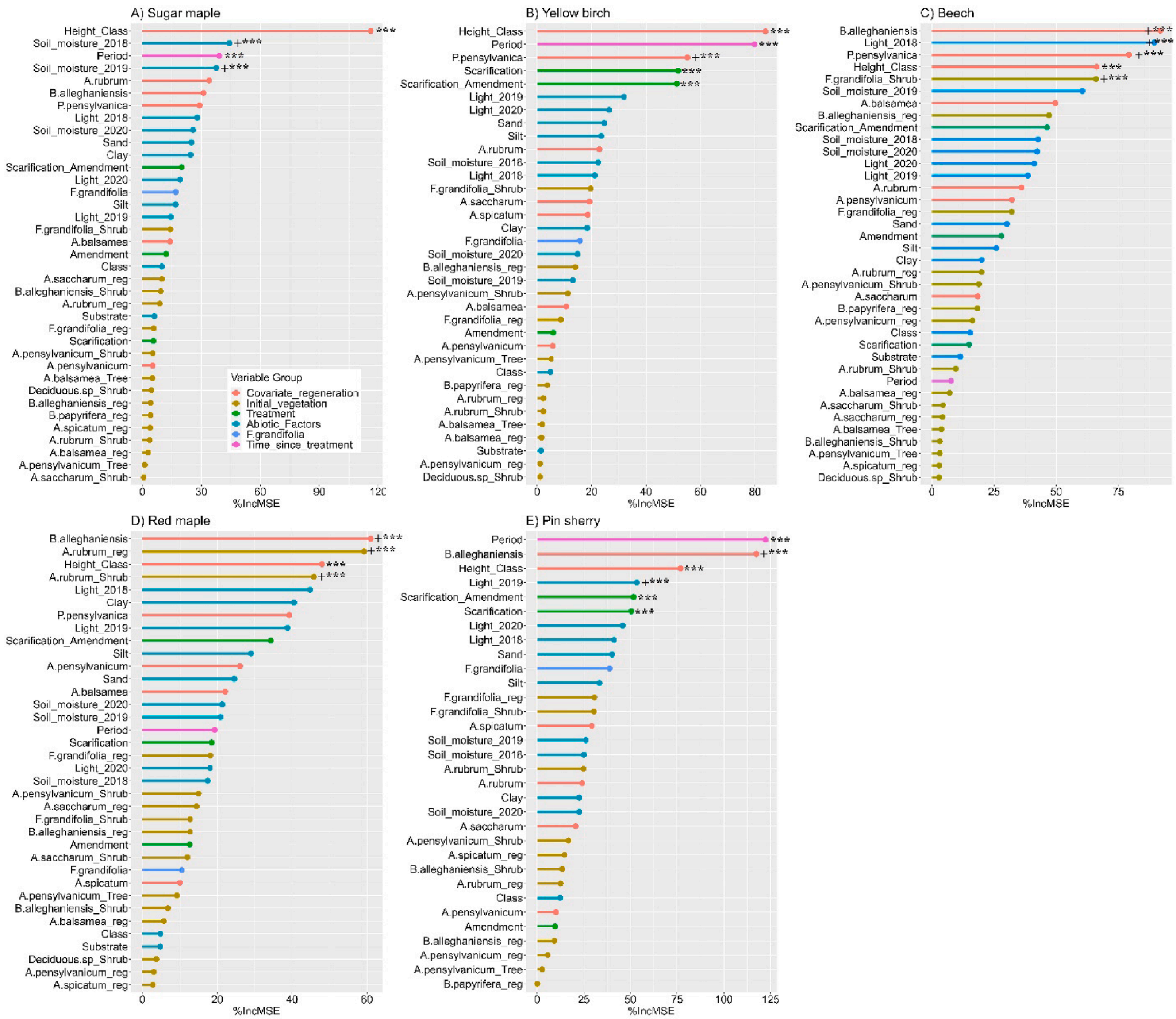

Fig. 8. Random forest analysis showing the influence of multiple variables/drivers on species seedling establishment. The percentage (%) decrease in the squared correlation coefficient (MSE) when a given variable was not included in the random forest model. Asterisks indicate that the variable is a significant driver at p = 0.05. The positive sign indicates a positive relationship between the variable and species/seedlings establishment.

Data availability

Data will be made available on request.

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David Paré: Conceptualization, Validation, Writing — original draft, Writing — review & editing.

Jérôme Laganière: Conceptualization, Funding acquisition, Project administration, Supervision, Methodology, Writing — original draft, Writing — review & editing.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foreco.2023.121071.

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